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Professional Summary

B.Tech Computer Science student focused on building production-oriented **Agentic AI systems, LLM applications, RAG pipelines, and multi-agent workflows**. Hands-on experience with Python, LangChain, LangGraph, FastAPI, vector databases, prompt engineering, asynchronous streaming, and AI agent orchestration. Strong foundation in Data Structures and Algorithms with experience developing end-to-end AI applications and backend APIs.

Technical Skills

- **Programming:** Python, Java, Dart, SQL
- **AI/ML:** Generative AI, LLMs, Prompt Engineering, RAG, Agentic AI, AI Agents, Multi-Agent Systems, Tool Calling, Structured Outputs
- **AI Frameworks:** LangChain, LangGraph, OpenAI SDK
- **Backend:** FastAPI, REST APIs, AsyncIO, SSE/Streaming APIs
- **Databases:** PostgreSQL, MySQL, ChromaDB, Qdrant
- **Core Concepts:** Embeddings, Vector Search, Semantic Search, Context Management, Short-Term/Long-Term Memory, Agent Orchestration, Workflow State Management
- **Development:** Git, GitHub, Docker, Linux, Pytest, Virtual Environments
- **Other:** Data Structures & Algorithms, OOP, System Design Fundamentals

Projects

Production-Grade Multi-Agent AI Framework *Python, LLMs, Agent Orchestration, AsyncIO*

- Designed a modular multi-agent framework in pure Python for building and orchestrating specialized AI agents.
- Implemented agent execution, task delegation, shared state management, memory handling, and workflow coordination.
- Designed reusable components for agent configuration, execution, tool integration, and response processing.
- Applied asynchronous programming patterns to support scalable agent execution.
- Focused on clean architecture, modularity, error handling, and extensibility for production-oriented AI applications.

Resume Analyzer & ATS Optimization Agent *Python, FastAPI, LangGraph, LangChain, LLMs*

- Built an AI-powered resume analysis workflow that evaluates resumes against job descriptions.
- Implemented a multi-stage LangGraph workflow involving resume analysis, interviewing, rewriting, and critic/evaluation agents.
- Designed conditional workflow routing to determine when additional resume rewriting or refinement is required.

- Developed FastAPI endpoints for interacting with the AI workflow.
- Implemented asynchronous workflow execution and streaming responses using SSE.
- Added structured state management and exception handling for reliable workflow execution.

Enterprise Knowledge Base RAG Platform

Python, FastAPI, LangChain, PostgreSQL, Vector Database

- Designed a production-oriented RAG platform for ingesting and querying enterprise documents.
- Implemented document processing, text chunking, embeddings, vector search, and context retrieval concepts.
- Designed PostgreSQL-backed metadata and document storage architecture.
- Planned support for ChromaDB/Qdrant for semantic vector retrieval.
- Implemented citation-aware retrieval to improve answer traceability and reduce unsupported responses.
- Designed REST APIs for document upload, indexing, semantic search, and question answering.

AI Blog Writer Agent

Python, LLM, Prompt Engineering, Agent Workflow

- Developed an LLM-powered content generation agent using structured prompting techniques.
- Applied the COSTAR prompting framework to improve consistency and quality of generated content.
- Designed workflows for generating, refining, and evaluating AI-generated articles.
- Implemented environment-based API configuration and error handling for external LLM APIs.

Practical Experience

Generative AI & Agentic AI Development

Self-Directed Projects

- Continuously developed hands-on projects involving LLM applications, RAG, AI agents, and agent orchestration.
- Worked with LangChain and LangGraph to build stateful and conditional AI workflows.
- Explored agent architectures including ReAct, tool-using agents, multi-agent collaboration, reflection, and self-refinement.
- Implemented asynchronous AI workflows using Python `asyncio`, `astream`, and Server-Sent Events.
- Worked with LLM APIs, embeddings, vector databases, prompt engineering, and structured outputs.
- Debugged API rate limits, asynchronous workflow failures, dependency conflicts, and backend integration issues.

Education

Bachelor of Technology — Computer Science and Engineering

Currently pursuing

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Software Engineering